pseudocode

1. minimize $f(x) = (w_1 * \text{passenger_waiting_Time}) + (w_2 * \text{operational_cost})$

w_1 = passenger Hapiness weight
w_2 = company savings weight
$w_1 + w_2 = 1$

2. Set No of particles → N
Set Inertia weight → w
Set cognitive coefficient → c_1 and
Social coefficient → c_2
Set Maximum Iterations T
set t = 0 (Literation counter)

3. For each particle i from 1 to N:
Initialize random position (x_i)
Initialize random velocity (v_i)
$pbest_i = x_i$

4. For each particle i from 1 to N:
Evaluate fitness: $f(x_i)$
g.best = position of particle with minimum $f(x_i)$

5. while $t < T$
for each particle i from 1 to N:
$V_i = (w \times V_i) + (c_1 * r_1 * (pbest_i - x_i)) + (c_2 * r_2 * (gbest - x_i))$

$x_i = x_i + V_i$

Sort(x_i.Departure_Time)
Round(x_i.vehicle_Assignment)
for each particle i from 1 to N:
Evaluate current_fitness = $f(x_i)$
if current_fitness < $f(pbest_i)$:
$pbest_i = x_i$

if current fitness < f(gbest):
gbest = x_i

t = t + 1

7. output gbest (global best position)
out f(gbest) (Best objective fun value).

output

Trips : 10
Max Buses : 6
particles : 50
Iterations : 150
$w_1 = 0.3$
$w_2 = 0.7$

--- RESUTS ---
Best score : 700.804
Times [0.1, 10.1, 20.0, 30.0, 39.9,
49.7, 59.9, 70.1, 80.1, 90.0]
Buses : [4, 6, 16, 4, 2, 3, 6, 6, 4, 3]

dp seen.
Sixth S
16/10/26.